

```
In [1]: print("01-04-2026")
```

```
01-04-2026
```

```
In [ ]:
```

```
In [2]: print("Introduction to Python Programming")
```

```
Introduction to Python Programming
```

```
In [24]: import pandas as pd
df = pd.read_excel(r"C:\Users\Jagatjyoti\Jagat_Code\31-03-2026\data.xlsx")
print(df.head())
```

	CountryName	CountryCode	BirthRate	InternetUsers	\
0	Aruba	ABW	10.244	78.9	
1	Afghanistan	AFG	35.253	5.9	
2	Angola	AGO	45.985	19.1	
3	Albania	ALB	12.877	57.2	
4	United Arab Emirates	ARE	11.044	88.0	

	IncomeGroup
0	High income
1	Low income
2	Upper middle income
3	Upper middle income
4	High income

```
In [5]: df.columns
```

```
Out[5]: Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
              'IncomeGroup'],
              dtype='object')
```

```
In [7]: df['CountryCode']
```

```
Out[7]: 0      ABW
1      AFG
2      AGO
3      ALB
4      ARE
...
190    YEM
191    ZAF
192    COD
193    ZMB
194    ZWE
Name: CountryCode, Length: 195, dtype: object
```

```
In [8]: df[['CountryCode', 'CountryName']]
```

Out[8]:

	CountryCode	CountryName
0	ABW	Aruba
1	AFG	Afghanistan
2	AGO	Angola
3	ALB	Albania
4	ARE	United Arab Emirates
...	...	...
190	YEM	Yemen, Rep.
191	ZAF	South Africa
192	COD	Congo, Dem. Rep.
193	ZMB	Zambia
194	ZWE	Zimbabwe

195 rows × 2 columns

In [9]: `df.head(1)`

Out[9]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income

In [11]: `df_cat = df[['CountryName', 'CountryCode', 'IncomeGroup']]`
`df_cat.head(1)`

Out[11]:

	CountryName	CountryCode	IncomeGroup
0	Aruba	ABW	High income

In [19]: `df_num = df[['BirthRate', 'InternetUsers']]`
`df_num.head(1)`

Out[19]:

	BirthRate	InternetUsers
0	10.244	78.9

In [21]: `df.describe`

```
Out[21]: <bound method NDFrame.describe of
         InternetUsers \
0          Aruba          ABW      10.244      78.9
1      Afghanistan      AFG      35.253       5.9
2          Angola      AGO      45.985      19.1
3          Albania      ALB      12.877      57.2
4  United Arab Emirates      ARE      11.044      88.0
..          ...          ...          ...
190      Yemen, Rep.      YEM      32.947      20.0
191      South Africa      ZAF      20.850      46.5
192      Congo, Dem. Rep.      COD      42.394       2.2
193          Zambia      ZMB      40.471      15.4
194          Zimbabwe      ZWE      35.715      18.5

         IncomeGroup
0          High income
1          Low income
2  Upper middle income
3  Upper middle income
4          High income
..          ...
190  Lower middle income
191  Upper middle income
192          Low income
193  Lower middle income
194          Low income

[195 rows x 5 columns]>
```

```
In [22]: df_num.describe
```

```
Out[22]: <bound method NDFrame.describe of      BirthRate  InternetUsers
0      10.244      78.9
1      35.253       5.9
2      45.985      19.1
3      12.877      57.2
4      11.044      88.0
..          ...          ...
190      32.947      20.0
191      20.850      46.5
192      42.394       2.2
193      40.471      15.4
194      35.715      18.5

[195 rows x 2 columns]>
```

```
In [25]: df.describe
```

```

Out[25]: <bound method NDFrame.describe of
         InternetUsers \
0          Aruba          ABW      10.244      78.9
1      Afghanistan      AFG      35.253       5.9
2          Angola      AGO      45.985      19.1
3          Albania      ALB      12.877      57.2
4  United Arab Emirates      ARE      11.044      88.0
..          ...          ...          ...
190      Yemen, Rep.      YEM      32.947      20.0
191      South Africa      ZAF      20.850      46.5
192      Congo, Dem. Rep.      COD      42.394       2.2
193          Zambia      ZMB      40.471      15.4
194          Zimbabwe      ZWE      35.715      18.5

         IncomeGroup
0          High income
1          Low income
2  Upper middle income
3  Upper middle income
4          High income
..          ...
190  Lower middle income
191  Upper middle income
192          Low income
193  Lower middle income
194          Low income

[195 rows x 5 columns]>

```

```
In [26]: df.describe()
```

```

Out[26]:

```

	BirthRate	InternetUsers
count	195.000000	195.000000
mean	21.469928	42.076471
std	10.605467	29.030788
min	7.900000	0.900000
25%	12.120500	14.520000
50%	19.680000	41.000000
75%	29.759500	66.225000
max	49.661000	96.546800

```
In [27]: df_num.describe()
```

Out[27]:

	BirthRate	InternetUsers
count	195.000000	195.000000
mean	21.469928	42.076471
std	10.605467	29.030788
min	7.900000	0.900000
25%	12.120500	14.520000
50%	19.680000	41.000000
75%	29.759500	66.225000
max	49.661000	96.546800

In [28]: `df_cat.describe()`

Out[28]:

	CountryName	CountryCode	IncomeGroup
count	195	195	195
unique	195	195	4
top	Aruba	ABW	High income
freq	1	1	67

In [29]: `df`

Out[29]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [31]: `df['InternetUsers'] < 2`

Out[31]:

```

0      False
1      False
2      False
3      False
4      False
...
190     False
191     False
192     False
193     False
194     False
Name: InternetUsers, Length: 195, dtype: bool

```

In [32]: `df[df['InternetUsers'] < 2]`

Out[32]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
11	Burundi	BDI	44.151	1.3	Low income
52	Eritrea	ERI	34.800	0.9	Low income
55	Ethiopia	ETH	32.925	1.9	Low income
64	Guinea	GIN	37.337	1.6	Low income
117	Myanmar	MMR	18.119	1.6	Lower middle income
127	Niger	NER	49.661	1.7	Low income
154	Sierra Leone	SLE	36.729	1.7	Low income
156	Somalia	SOM	43.891	1.5	Low income
172	Timor-Leste	TLS	35.755	1.1	Lower middle income

In [37]: `df[df['BirthRate'] > 40]`

Out[37]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
2	Angola	AGO	45.985	19.1	Upper middle income
11	Burundi	BDI	44.151	1.3	Low income
14	Burkina Faso	BFA	40.551	9.1	Low income
65	Gambia, The	GMB	42.525	14.0	Low income
115	Mali	MLI	44.138	3.5	Low income
127	Niger	NER	49.661	1.7	Low income
128	Nigeria	NGA	40.045	38.0	Lower middle income
156	Somalia	SOM	43.891	1.5	Low income
167	Chad	TCD	45.745	2.3	Low income
178	Uganda	UGA	43.474	16.2	Low income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income

In [34]: `len(df[df['InternetUsers'] < 2])`

Out[34]: 9

In [35]: `df[df['BirthRate'] > 40]`

Out[35]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
2	Angola	AGO	45.985	19.1	Upper middle income
11	Burundi	BDI	44.151	1.3	Low income
14	Burkina Faso	BFA	40.551	9.1	Low income
65	Gambia, The	GMB	42.525	14.0	Low income
115	Mali	MLI	44.138	3.5	Low income
127	Niger	NER	49.661	1.7	Low income
128	Nigeria	NGA	40.045	38.0	Lower middle income
156	Somalia	SOM	43.891	1.5	Low income
167	Chad	TCD	45.745	2.3	Low income
178	Uganda	UGA	43.474	16.2	Low income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income

In [36]: `len(df[df['BirthRate'] > 40])`

Out[36]: 12

In [40]: `len( df[(df['BirthRate'] > 40) & (df['InternetUsers']<2)] )`

Out[40]: 3

In [42]: `df[(df['BirthRate'] > 40) & (df['InternetUsers']<2)]`

Out[42]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
11	Burundi	BDI	44.151	1.3	Low income
127	Niger	NER	49.661	1.7	Low income
156	Somalia	SOM	43.891	1.5	Low income

In [44]: `df[(df['IncomeGroup'] == 'High income') ]`

Out[44]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.90	High income
4	United Arab Emirates	ARE	11.044	88.00	High income
5	Argentina	ARG	17.716	59.90	High income
7	Antigua and Barbuda	ATG	16.447	63.40	High income
8	Australia	AUS	13.200	83.00	High income
...	...	...	...	...	...
174	Trinidad and Tobago	TTO	14.590	63.80	High income
180	Uruguay	URY	14.374	57.69	High income
181	United States	USA	12.500	84.20	High income
184	Venezuela, RB	VEN	19.842	54.90	High income
185	Virgin Islands (U.S.)	VIR	10.700	45.30	High income

67 rows × 5 columns

In [45]:

```
df[(df['IncomeGroup'] == 'Low income') ]
```

Out[45]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
1	Afghanistan	AFG	35.253	5.90	Low income
11	Burundi	BDI	44.151	1.30	Low income
13	Benin	BEN	36.440	4.90	Low income
14	Burkina Faso	BFA	40.551	9.10	Low income
29	Central African Republic	CAF	34.076	3.50	Low income
38	Comoros	COM	34.326	6.50	Low income
52	Eritrea	ERI	34.800	0.90	Low income
55	Ethiopia	ETH	32.925	1.90	Low income
64	Guinea	GIN	37.337	1.60	Low income
65	Gambia, The	GMB	42.525	14.00	Low income
66	Guinea-Bissau	GNB	37.503	3.10	Low income
77	Haiti	HTI	25.345	10.60	Low income
93	Cambodia	KHM	24.462	6.80	Low income
99	Liberia	LBR	35.521	3.20	Low income
111	Madagascar	MDG	34.686	3.00	Low income
115	Mali	MLI	44.138	3.50	Low income
120	Mozambique	MOZ	39.705	5.40	Low income
123	Malawi	MWI	39.459	5.05	Low income
127	Niger	NER	49.661	1.70	Low income
132	Nepal	NPL	20.923	13.30	Low income
148	Rwanda	RWA	32.689	9.00	Low income
154	Sierra Leone	SLE	36.729	1.70	Low income
156	Somalia	SOM	43.891	1.50	Low income
158	South Sudan	SSD	37.126	14.10	Low income
167	Chad	TCD	45.745	2.30	Low income
168	Togo	TGO	36.080	4.50	Low income
177	Tanzania	TZA	39.518	4.40	Low income
178	Uganda	UGA	43.474	16.20	Low income
192	Congo, Dem. Rep.	COD	42.394	2.20	Low income
194	Zimbabwe	ZWE	35.715	18.50	Low income

In [47]: `df['IncomeGroup'].unique()`

```
Out[47]: array(['High income', 'Low income', 'Upper middle income',
               'Lower middle income'], dtype=object)
```

```
In [48]: df['IncomeGroup'].nunique()
```

```
Out[48]: 4
```

```
In [49]: df.columns
```

```
Out[49]: Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
               'IncomeGroup'],
               dtype='object')
```

```
In [50]: df.columns = ['a', 'b', 'c', 'd', 'e']
```

```
In [51]: df.columns
```

```
Out[51]: Index(['a', 'b', 'c', 'd', 'e'], dtype='object')
```

```
In [52]: df.columns = ['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
               'IncomeGroup']
```

```
In [53]: df.columns
```

```
Out[53]: Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
               'IncomeGroup'],
               dtype='object')
```

```
In [54]: df.head(2)
```

```
Out[54]:
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income

```
In [59]: df['mycalc'] = df['BirthRate'] * df['InternetUsers']
```

```
In [58]:
```

```
In [60]: df.head()
```

```
Out[60]:
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup	mycalc
0	Aruba	ABW	10.244	78.9	High income	808.2516
1	Afghanistan	AFG	35.253	5.9	Low income	207.9927
2	Angola	AGO	45.985	19.1	Upper middle income	878.3135
3	Albania	ALB	12.877	57.2	Upper middle income	736.5644
4	United Arab Emirates	ARE	11.044	88.0	High income	971.8720

```
In [61]: df=df.drop('mycalc', axis=1)
```

```
In [62]: df.head()
```

```
Out[62]:
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

```
In [64]: df=df.drop('mycalc', axis=0) # row dont have a mycalc value
```

```

-----
KeyError                                Traceback (most recent call last)
Cell In[64], line 1
----> 1 df=df.drop('mycalc', axis=0)

File ~\anaconda3\Lib\site-packages\pandas\core\frame.py:5344, in DataFrame.drop(self, labels, axis, index, columns, level, inplace, errors)
    5196 def drop(
    5197     self,
    5198     labels: IndexLabel | None = None,
    (... )
    5205     errors: IgnoreRaise = "raise",
    5206 ) -> DataFrame | None:
    5207     """
    5208     Drop specified labels from rows or columns.
    5209     (...)
    5342         weight  1.0      0.8
    5343     """
-> 5344     return super().drop(
    5345         labels=labels,
    5346         axis=axis,
    5347         index=index,
    5348         columns=columns,
    5349         level=level,
    5350         inplace=inplace,
    5351         errors=errors,
    5352     )

File ~\anaconda3\Lib\site-packages\pandas\core\generic.py:4711, in NDFrame.drop(self, labels, axis, index, columns, level, inplace, errors)
    4709 for axis, labels in axes.items():
    4710     if labels is not None:
-> 4711         obj = obj._drop_axis(labels, axis, level=level, errors=errors)
    4713 if inplace:
    4714     self._update_inplace(obj)

File ~\anaconda3\Lib\site-packages\pandas\core\generic.py:4753, in NDFrame._drop_axis(self, labels, axis, level, errors, only_slice)
    4751     new_axis = axis.drop(labels, level=level, errors=errors)
    4752     else:
-> 4753     new_axis = axis.drop(labels, errors=errors)
    4754     indexer = axis.get_indexer(new_axis)
    4756 # Case for non-unique axis
    4757 else:

File ~\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:7000, in Index.drop(self, labels, errors)
    6998 if mask.any():
    6999     if errors != "ignore":
-> 7000         raise KeyError(f"{labels[mask].tolist()} not found in axis")
    7001     indexer = indexer[~mask]
    7002     return self.delete(indexer)

KeyError: "[ 'mycalc' ] not found in axis"

```

In []: